Factor-base shape search at $m \geq 3$: goals G1, G2, G3, G5

Lane `shape-search`, 2026-10-15 — stage diagnostic, toy binary fields, no S, no speedup

Sources. Goals frozen in commit `110117d8c` of `research/notes/index-calculus/RESEARCH_FACTOR_BASE_SHAPE_SEARCH.md` §7. Protocol `research/shape_search_g1_g6_20261015/PROTOCOL.md`, committed in `3bdf9b06e` before any measured cell. Instrument `examples/factor_base_shape_ladder.rs` (chained-S_3 ladder, closure-sweep shape builders, class-0 targets $\mathrm{Tr}(1/x_R) = 0$, $b = 1$). Seed `20261015`. Records: `runs/*.jsonl`; scores: `summary.md`, written by the Rust `score` subcommand. Host: four-core container shared with four other lanes; `nice 10`, one thread, 2.5 GB address cap.

1. Verdicts under the frozen kill conditions

goal	kill condition	verdict (scope)
G1	no cell with the gap	**killed on the scaled cells** $(5,2)$, $(7,3)$, $(11,4)$: gp, fp, coord gap 0.000 in every cell. Frozen cells $(13,5)$, $(17,6)$, $(19,7)$ **not testable** (resource).
G2	rise ≥ 1 vs plain chain	**survives at these sizes**: rise 0.000 in every arm ($s \in \{1, 2, l\}$; second and ordered; random and fp) at $(5,2)$, $(7,3)$, $(9,3)$.
G3	class-0 D flat in the panel to 0.3	**killed**: spread 0.000 at $(5,2)$, $(7,3)$, $(11,4)$ (6, 7, 5 moduli).
G4	—	not built (budget).
G5	class-0 D^* flat in β to 0.3	**killed** (expected null): β-spread 0.06–0.19 at $m = 2$, $(7,3)$–$(13,6)$.
G6	—	not run: no survivor of G1 or G3.

Positive control: the subfield $\mathbb{F}_8 \subset \mathbb{F}_{2^9}$ refutes at 5 where random reads 6 at $(9,3)$ — the instrument sees a one-degree shape effect at $m = 3$. Negative control (`random2` vs `random`): gap 0.000 in every cell.

2. Measured values (class-0 mean refutation degree; draws in brackets)

cell	random	random2	gp	fp	coord	subfield	unknowns
$(5,2)$	5.000 (8)	5.000 (8)	5.000 (8)	5.000 (8)	5.000 (7)	—	11
$(7,3)$	6.000 (8)	6.000 (8)	6.000 (8)	6.000 (8)	6.000 (8)	—	16
$(9,3)$ control	6.000 (8)	—	—	—	—	**5.000** (8)	18
$(11,4)$	6.000 (6)	6.000 (6)	6.000 (6)	6.000 (6)	6.000 (6)	—	23

Every draw refuted; none censored or capped; within each cell every draw reads the same degree. The $(13,5)$ feasibility probe (one random draw, $d_{\max} = 6$) did not finish in 60 min and was stopped; $(17,6)$ and $(19,7)$ (35 and 40 unknowns) are out of budget by derivation.

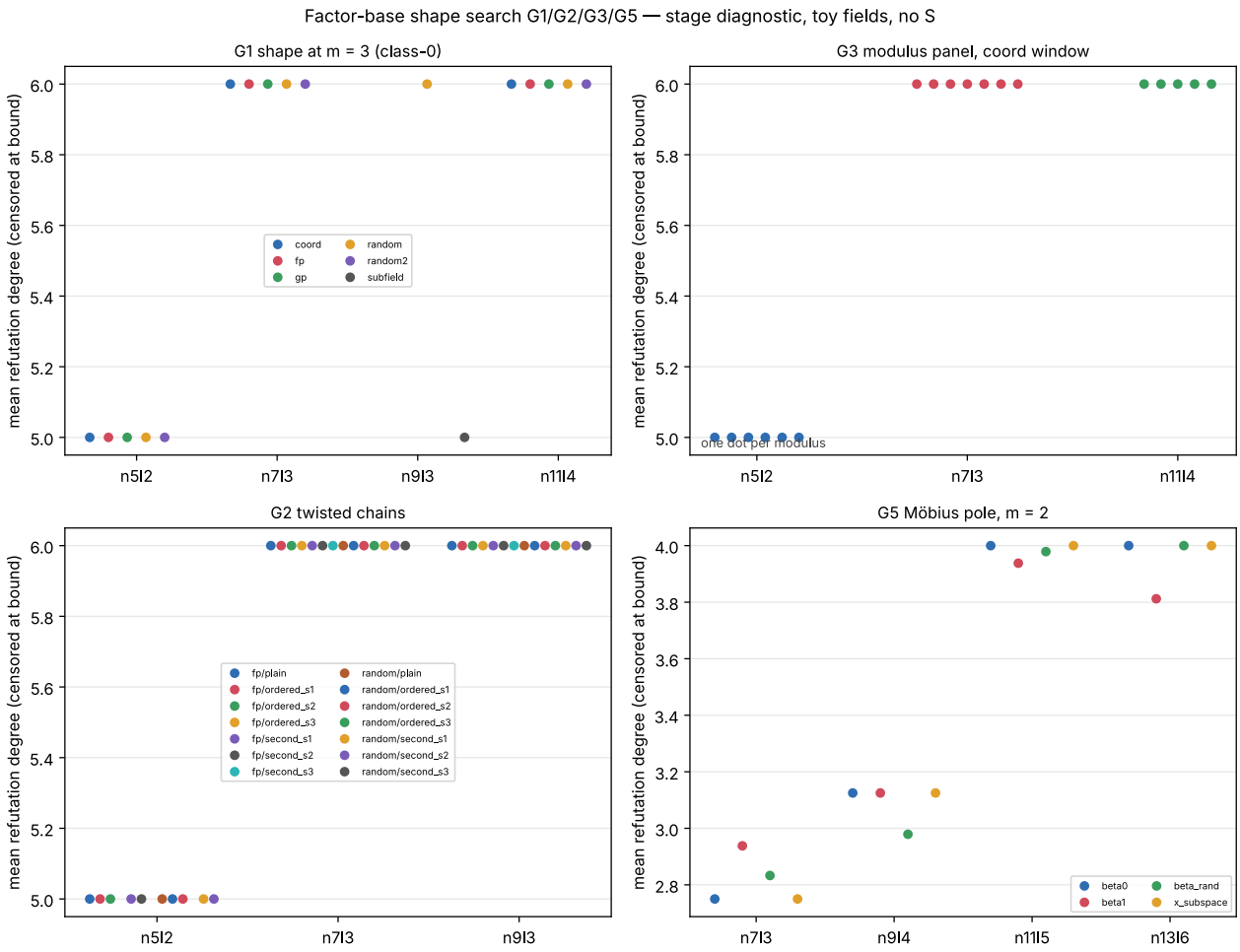


Figure 1: Class-0 mean refutation degree per arm and cell (values from `summary.md`; seed 20261015). G3: one dot per modulus. G5: $m = 2$. Censored counts would be annotated; there are none.

3. Notes

- **G2 yield (derived).** Random bases: twisted arms decompose more class-0 targets than the plain chain (e.g. $(7, 3)$: 0.571 plain vs 0.952 ordered, $s = 1$); the small fields saturate, so the hypothesised factor $m! = 6$ cannot be read. fp bases at $(5, 2)$ and $(7, 3)$ saw identical satisfiable/unsatisfiable sequences in every arm — observed on a few dozen targets, unexplained.
- **G5.** $\beta = 0$ reproduces the plain x-subspace arm draw for draw: at $b = 1$, $(u_1 u_2)^2 S_3(1/u_1, 1/u_2, x_3) = S_3(u_1, u_2, x_3)$ (derived). $\beta = 1$ (A14) is lowest at $(13, 6)$ by 0.19 (3 draws of 16 at 3), under the threshold. The frozen `coordinate_quotients.rs` engine measures no refutation degree; the descended pole system was substituted (disclosed).
- **Reading.** At these sizes the degree is set by l alone (5 at $l = 2$; 6 at $l = 3, 4$), matching `RESEARCH_DREG_MEASUREMENT.md` Result 10. Shape, modulus and twist move it by zero. The subfield moves it by one but has class-0 yield 0 at $b = 1$ (a subgroup) and does not exist at prime n. The degree is quantized and constant within a cell, so the open question is $l \geq 5$, out of reach here.
- **Note §5, L239.** The frozen scale-8 run never landed: `experiments/factor_base_closure_sweep.json` is absent from `origin/main` and no commit ever touched it; §5 is still "Pending".
- **Build provenance.** Records were made with a shared-target build; a lane-private rebuild re-ran G1 $(5, 2), (7, 3), (9, 3)$, G2 $(5, 2), (7, 3)$, G3 $(5, 2), (7, 3)$, all of G5 and the G1 $(11, 4)$ gp arm: 718 measured lines, all identical apart from wall time (`crosscheck.out`).
- Nothing here bears on ECC2K-130.